DX10 ON A BUDGET

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Last months' graphics card test was all about raw speed, blistering frame rates and pixel crunching power. Those cards were suitable for those looking at premium graphics performance and were willing to shell out for it. There's an adage – you pay for what you get. So hardcore gamers and anyone wanting a blazing fast video solution, whether for winning benchmarks or bragging rights, would be shopping for a GeForce GTX 280 or a Radeon HD 4870. Those looking for something cheaper would make do the GeForce 9800 GTX range of cards. But what about the casual gamer, the HTPC user or someone who isn't interested in gaming at all, but wants an entry level graphics card that speeds up his PC in general?

It's for you that we've conducted this test. Far away from 8x Antialiasing and *Crysis* there exists a world of moderation. Where PC users are looking to upgrade from bottleneck riddled integrated graphics to a better multimedia experience, the kind that can only arise from using a discrete graphics card. As you no doubt know, not only does a graphics card free up CPU utilisation and memory utilisation by adding a discrete GPU and video memory to the equation, but also aids any kind of multimedia activity. In fact, a discrete GPU will benefit your PCs performance more than a CPU that is say 50 per cent costlier. So if you have the choice of spending Rs 15,000 on a fast CPU or Rs 10,000 on a moderately fast CPU and Rs 5,000 on a graphics card, you should ideally opt for the latter. If we could clear one misconception it would have to be this arcane concept that 70 per cent of PC users have, that is a graphics card is solely for gamers. In India this percentage is more like 90 per cent. There was a time when graphics cards were ridiculously priced; to the extent that anybody shopping for a PC for anything other than gaming would drop the idea totally. While the Indian prices aren't on par with US rates yet, we've noticed somewhat of an equalisation effect in force. Prices usually drop a few months after a product launch. This time round we've seen something that hasn't occurred in the past.

So you want DX 10 cheap eh? One of these will satisfy

For the first time, prices of all graphics cards in general have fallen. Earlier a high-end card used to cost around the Rs 30,000 mark. Now with the entry of the Radeon HD 4870 and the GTX 260, truly powerful cards are available for as little as Rs 18,000 to Rs 25,000. And these aren't after price cuts; these new products debuted at these prices! Due to this, the mid-range now consists of some juicy offerings at throw-away prices. The GeForce 9800 GTX is a good example. The mid-range is now the Rs 9,000 to Rs 15,000 mark—with a few exceptions. As a result the entry level has got even juicier. Another factor contributing to the value existing in this segment is the performance of the entry level options. Previously the phrase "entry level" was used in disdain by anyone even remotely interested in gaming, and with cards such as the GeForce 7300 GS and Radeon X1300 SE hovering around, we couldn't fault people thinking on these lines. After all, adding a graphics card should do *something* for graphics, right?

This time around, the entry level segment contains some meaty options for gamers in the form of NVIDIA's GeForce 9600 GT / GSO GPUs and ATI's Radeon 3850. The GeForce 9500 GT and 9400 GT bring up the rear for NVIDIA, while ATI has the Radeon 3650 cards. ATI did send us a Radeon 4650 and 4550 to test, but for some reason we could not get these cards to work under DX 10 – we feel this is a driver issue, and the latest version of Catalyst (8.9) was unable to work these cards with 3D Mark Vantage or *Crysis*. Therefore we regret to inform you that we had to exclude these cards from the test. But look out for a *Bazaar* review as soon as we can get our hands on a working driver.

GeForce 9600 GT / GSO

Shoestring performers

Ever since its launch the 9600 GT has been a winner. With just 64 stream processors (SPs) this GPU featured a 256-bit GDDR3 memory interface and could actually match the 112 SP bearing GeForce 8800 GT. This GPU is based around a 65 nm fabrication process which means any card based on this GPU will run quite cool and a single slot cooling solution will suffice. The 9600 GSO is a weird part. Its specifications cause us to wonder what exactly NVIDIA was thinking. On one hand they gave it 96 SPs – a bump up from the 9600 GT. On the other hand the memory bus width was butchered from 256-bits to 192-bits. This doesn't seem to make sense, as both these changes are counter-productive of each other. Since both these cores aren't too power hungry, they need a single six pin power connector.

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Features

There were three 9600 GT cards and one 9600 GSO. Palit sent us both a 9600 GT and a 9600 GSO. Both cards had custom cooling solutions that really kept temperatures below what the stock cooling can achieve. Although these cooling solutions do take up an additional slot; making these cards dual slot solutions.